PRACTICAL NO : 3

Name : Ghanwat Payal Ganesh Roll No : 222 Class : SE (AIML) Batch : S1

Problem Statement: Circular Queue-- CO1, CO2

Implement Circular Queue using Array. Perform following operations on it. a. Insertion (Enqueue) b. Deletion (Dequeue) c. Display (Note: Handle queue full condition by considering a fixed size of a queue.)

Code :

```
#include   <iostream>

using namespace std;


#define SIZE 5


class CircularQueue

{

 int queue[SIZE];

 int front, rear;


public:

 CircularQueue()

 {

  front = -1;

  rear = -1;

}

// Enqueue

void enqueue(int value)

{

 if ((rear + 1) % SIZE == front)

 {

  cout << "Queue is Full!" << endl;
```

```
 return;
}

if (front == -1)
 front = 0;

rear = (rear + 1) % SIZE;
queue[rear] = value;

cout << value << " inserted successfully." << endl;
}
// Dequeue
void dequeue()
{
 if (front == -1)
 {
  cout << "Queue is Empty!" << endl;
    return;
 }
  cout << queue[front] << " deleted Successfully." << endl;

 if (front == rear)
 {
  front = rear = -1;
 }
  else
  {
   front = (front + 1) % SIZE;
```

```
  }
}

// Display
void display()
{
 if (front == -1)
 {
  cout << "Queue is Empty!" << endl;
  return;
 }

 cout << "Circular Queue: ";

 int i = front;
 while (true)
{
 cout << queue[i] << " ";

 if (i == rear)
   break;

 i = (i + 1) % SIZE;
 }
  cout << endl;
  }
};
```

```
int main()
{
 CircularQueue q;
 int choice, value;

 do
{
 cout << "\n----- Circular Queue Menu----- " << endl;
 cout << "1. Enqueue" << endl;
 cout << "2. Dequeue" << endl;
 cout << "3. Display" << endl;
 cout << "4. Exit" << endl;

 cout << "Enter your choice: ";
 cin >> choice;

 switch (choice)
 {
 case 1:
   cout << "Enter value: ";
   cin >> value;
   q.enqueue(value);
   break;
 case 2:
   q.dequeue();
   break;
 case 3:
   q.display();
```

```
  break;
 case 4:
  cout << "Program Exited." << endl;
  break;
 default:
  cout << "Invalid Choice!" << endl;
 }
 } while (choice != 4);
 return 0;
}
```

Output:

```
----- Circular Queue Menu -----
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
Queue is Empty!

----- Circular Queue Menu -----
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter value: 20
20 inserted successfully.
```

----- Circular Queue Menu -----

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 3

Circular Queue: 20

----- Circular Queue Menu -----

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 10

Invalid Choice!

----- Circular Queue Menu -----

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 4

Program Exited.

=== Code Execution Successful ===